

A decreasing-power Orlicz family outside the class $\mathcal{X}$

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Source and target

The source paper is Astashkin–Lykov–Milman, *Majorization revisited: Comparison of norms in interpolation scales*, arXiv:2107.11854 [1]. In Section 6 the authors define the class $\mathcal{X}$ of rearrangement invariant spaces X on $(0, \infty)$ with the following property. If a pair (f, g) satisfies the Nazarov–Podkorytov single-crossing condition

$$f^*(t) \geq g^*(t) \quad (t \leq t_0), \quad f^*(t) \leq g^*(t) \quad (t \geq t_0),$$

and if $\|f\|_X \geq \|g\|_X$, then

$$\|f^* \chi_{(0,s)}\|_X \geq \|g^* \chi_{(0,s)}\|_X \quad \text{for every } s > 0.$$

They state that a complete characterization of $\mathcal{X}$ is open. They prove that Lorentz spaces $\Lambda_r(\varphi)$ belong to $\mathcal{X}$, that the Marcinkiewicz spaces $M(t^\alpha)$ do not, and that an Orlicz space L_M belongs to $\mathcal{X}$ under a monotonicity assumption on $M'(\varepsilon x)/M'(x)$.

Result

This packet gives a negative Orlicz-family classification complementary to the positive Orlicz theorem in the source paper.

Theorem 1. *Let $0 < p < q$. Define*

$$N_{p,q}(u) = \begin{cases} u^q, & 0 \leq u \leq 1, \\ u^p, & u \geq 1, \end{cases} \quad M_{p,q}(u) = \int_0^u N_{p,q}(v) dv.$$

Then $M_{p,q}$ is a Δ_2 Orlicz function, but the Orlicz space $L_{M_{p,q}}(0, \infty)$ equipped with the Luxemburg norm is not in $\mathcal{X}$.

Proof intuition

The source paper’s positive Orlicz theorem says, in effect, that scaling an Orlicz modular down by ε must preserve the direction of a single-crossing modular comparison. The family above has the opposite geometry: near zero the modular behaves like a higher power, while at infinity it behaves like a lower power. Thus scaling down suppresses the low-level part more strongly than the high-level part.

We exploit this by building a two-step prefix. On the prefix, f has a short high block and a long low block, while g has one intermediate block. At the larger modular scale this prefix favors g ; after

scaling by ε it favors f . A common low tail is then appended to both functions. The common tail does not affect the failed prefix comparison, but it normalizes the global Luxemburg comparison so that $\|f\| \geq \|g\|$.

A crossing-block criterion

We use the Luxemburg norm

$$\|h\|_{L_M} = \inf \left\{ \lambda > 0 : \int_0^\infty M(|h(t)|/\lambda) dt \leq 1 \right\}.$$

Lemma 1. *Let M be a strictly increasing Orlicz function. Suppose there are $0 < d < a < b$ and $0 < \varepsilon < 1$ such that*

$$\frac{M(\varepsilon a) - M(\varepsilon d)}{M(a) - M(d)} < \frac{M(\varepsilon b) - M(\varepsilon a)}{M(b) - M(a)}. \quad (1)$$

Then $L_M(0, \infty) \notin \mathcal{X}$.

Proof. Condition (1) is equivalent to the existence of a number $\theta > 0$ such that

$$\frac{M(b) - M(a)}{M(a) - M(d)} < \theta < \frac{M(\varepsilon b) - M(\varepsilon a)}{M(\varepsilon a) - M(\varepsilon d)}. \quad (2)$$

The left inequality gives

$$M(b) + \theta M(d) < (1 + \theta)M(a). \quad (3)$$

Choose $B > 0$ so that

$$B(M(b) + \theta M(d)) < 1 < B(1 + \theta)M(a), \quad (4)$$

and put $C = \theta B$ and $s_0 = B + C$.

On $(0, s_0)$ define decreasing step functions

$$U_0(t) = \begin{cases} b, & 0 < t < B, \\ d, & B < t < s_0, \end{cases} \quad V_0(t) = a \quad (0 < t < s_0).$$

By (4),

$$\int_0^{s_0} M(U_0(t)) dt < 1 < \int_0^{s_0} M(V_0(t)) dt. \quad (5)$$

The right inequality in (2) gives

$$BM(\varepsilon b) + CM(\varepsilon d) > (B + C)M(\varepsilon a). \quad (6)$$

Since $\varepsilon < 1$ and M is increasing, the left side of (6) is strictly smaller than $BM(b) + CM(d)$, hence is < 1 by (4). Therefore

$$T = \frac{1 - \int_0^{s_0} M(\varepsilon U_0(t)) dt}{M(\varepsilon d)}$$

is positive. Extend the unscaled functions by a common tail:

$$U(t) = \begin{cases} U_0(t), & 0 < t < s_0, \\ d, & s_0 < t < s_0 + T, \\ 0, & t > s_0 + T, \end{cases} \quad V(t) = \begin{cases} V_0(t), & 0 < t < s_0, \\ d, & s_0 < t < s_0 + T, \\ 0, & t > s_0 + T. \end{cases}$$

Finally set $f = \varepsilon U$ and $g = \varepsilon V$.

The functions are already decreasing rearrangements. With $t_0 = B$ they satisfy the source paper's NP condition: $f^* \geq g^*$ on $(0, B)$ and $f^* \leq g^*$ on (B, ∞) .

At the global scale $\lambda = 1$ we have, by the choice of T ,

$$\int_0^\infty M(f(t)) dt = 1,$$

while (6) implies

$$\int_0^\infty M(g(t)) dt = \int_0^{s_0} M(\varepsilon V_0(t)) dt + TM(\varepsilon d) < 1.$$

Thus $\|f\|_{L_M} = 1 \geq \|g\|_{L_M}$.

However, on the prefix $(0, s_0)$, (5) says

$$\int_0^{s_0} M(f(t)/\varepsilon) dt < 1 < \int_0^{s_0} M(g(t)/\varepsilon) dt.$$

Consequently

$$\|f^* \chi_{(0, s_0)}\|_{L_M} \leq \varepsilon < \|g^* \chi_{(0, s_0)}\|_{L_M}.$$

This contradicts the defining implication for $\mathcal{X}$, so $L_M \notin \mathcal{X}$. □

Proof of the theorem

The function $N_{p,q}$ is increasing and continuous, so $M_{p,q}$ is an Orlicz function. Its growth is polynomial at both zero and infinity, so $M_{p,q} \in \Delta_2$.

Fix any $0 < \varepsilon < 1$ and choose $0 < d < a < 1$. Since $\varepsilon a < 1$,

$$\frac{M_{p,q}(\varepsilon a) - M_{p,q}(\varepsilon d)}{M_{p,q}(a) - M_{p,q}(d)} = \varepsilon^{q+1}. \quad (7)$$

On the other hand, as $b \rightarrow \infty$,

$$\frac{M_{p,q}(\varepsilon b) - M_{p,q}(\varepsilon a)}{M_{p,q}(b) - M_{p,q}(a)} \longrightarrow \varepsilon^{p+1}. \quad (8)$$

Because $0 < p < q$ and $0 < \varepsilon < 1$, we have $\varepsilon^{q+1} < \varepsilon^{p+1}$. Thus, for all sufficiently large b , condition (1) holds. Lemma 1 applies and gives $L_{M_{p,q}} \notin \mathcal{X}$.

Concrete sanity check

The script `code/verify_concrete.py` checks one explicit instance: $p = 1$, $q = 2$, $\varepsilon = 1/2$, $d = 1/4$, $a = 1/2$, $b = 20$. It prints

$$0.125 = \frac{M(\varepsilon a) - M(\varepsilon d)}{M(a) - M(d)} < \frac{M(\varepsilon b) - M(\varepsilon a)}{M(b) - M(a)} \approx 0.2494,$$

then constructs admissible B, C, T and verifies the prefix and global modular inequalities used in the proof.

Limitations and novelty check

This is not a complete characterization of $\mathcal{X}$. It gives a new explicit negative family inside Orlicz spaces, complementing the source paper’s sufficient condition and positive examples. The result suggests that the monotonicity of the scaling profile is close to the correct Orlicz-side dividing line, but this packet proves only the negative direction for a broad piecewise-power family.

On 2026-07-03, web searches for exact and close phrases around *Majorization revisited*, $\mathcal{X}$, Orlicz spaces, Nazarov–Podkorytov, and the ratio $M'(\varepsilon x)/M'(x)$ found the source arXiv page but no later answer or matching Orlicz counterexample family. Novelty confidence is therefore moderate, pending expert literature review.

Human-review recommendation

The main point to check is Lemma 1: the functions must satisfy the exact NP crossing condition, the common tail must not enter the prefix comparison, and the Luxemburg modular normalizations must imply the stated norm inequalities. The specialization to $M_{p,q}$ is then elementary.

References

- [1] S. V. Astashkin, K. V. Lykov, and M. Milman, *Majorization revisited: Comparison of norms in interpolation scales*, arXiv:2107.11854, 2021.